

Danaboina Shanmukeshwar

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SUMMARY

Senior Software Development Engineer with 4+ years building distributed systems in Rust and Python. Designed 57 domain-specific WASM plugins (Extism PDK), multi-tenant platforms with hash-partitioned PostgreSQL, multi-region AWS infrastructure (Terraform/Terragrunt), and real-time voice AI pipelines. Specialized in backend architecture, systems programming, and infrastructure at scale.

SKILLS

- **Languages:** Rust, Python, TypeScript, SQL
- **Rust Ecosystem:** Tokio, WebAssembly, Extism, Cargo Workspaces
- **Backend & Databases:** FastAPI, PostgreSQL, Redis, Elasticsearch, Neo4j, Kafka
- **Cloud & DevOps:** AWS, Kubernetes, Terraform, Terragrunt, Docker, GitHub Actions
- **ML & Inference:** LLM Fine-tuning, QLoRA, ONNX, HuggingFace Transformers, PEFT
- **Architecture:** System Design, Distributed Systems, Microservices, Bitemporality

EXPERIENCE

Senior Software Development Engineer

Mar 2025 - Present

Arrowhead

Bangalore, India

- Designed 57 production Rust WASM plugins (Extism PDK) for a voice AI calling platform across insurance, pharma, diagnostics, and debt collection verticals — with pre-call data enrichment, live LLM tool-calling, and post-call processing hooks
- Built a Rust CLI toolchain (Tokio, Clap) for internal infrastructure operations, including a multi-region Kubernetes proxy that tunnels EKS, Aurora PostgreSQL, ElastiCache Redis, and VictoriaLogs across US and India environments
- Integrated 5 telephony providers with Temporal workflow orchestration for outbound voice campaign scheduling, call reconciliation, and disposition tracking
- Built real-time voice pipeline with Deepgram STT, ElevenLabs TTS, and Speechmatics; added smart turn detection and LLM-based PII redaction; orchestrates 10+ LLM providers via LiteLLM
- Fine-tuned LLMs (Gemma 3, MuRIL) for multilingual intent classification to enable predictive response caching in the voice pipeline, reducing perceived latency by pre-computing LLM responses before the user finishes speaking
- Designed multi-tenant API with hash-partitioned PostgreSQL, HMAC-SHA256 request signing with nonce-based replay attack prevention, and JWT blacklisting
- Scaled heartbeat-driven scheduler (picks and executes outbound calls) from 1,000 to 10,000 concurrent executions through async task queue redesign and worker pool tuning
- Built bi-temporal billing system tracking both valid-time and transaction-time, enabling accurate retroactive corrections and audit-safe charge history
- Implemented cell-based multi-tenant architecture isolating tenant workloads into independent cells for fault containment and horizontal scaling
- Provisioned India (ap-south-1) AWS infrastructure from scratch with Terraform/Terragrunt — EKS, Aurora PostgreSQL, ElastiCache Redis, ECR — deploying 6 microservices across dev and prod alongside the existing US region

Founding Engineer & Head of Technology

Jun 2024 - Mar 2025

AroundMe (Closed)

Hyderabad, India

- Founded and led team of 10 building location-based social app (65K+ downloads across Play Store and App Store)
- Architected distributed backend: FastAPI, PostgreSQL, Redis cluster (master + 3 slaves), Elasticsearch, Neo4j
- Implemented ML-powered recommendations and AI content moderation
- Kubernetes deployment with GitOps (ArgoCD) and GitHub Actions CI/CD

Software Development Engineer

Aug 2022 - May 2023

OpenCubicles Technologies

Hyderabad, India

- Built smart AC maintenance system with camera feeds, custom FTP server, and AWS Lambda pipeline for automated fault detection

- Developed Shopify Builder Plugin with React drag-and-drop interface and bidirectional cross-iframe communication
- Implemented Kafka event processing and Redis caching for real-time analytics

Software Development Engineer Intern

Jun 2021 - Aug 2022

OpenCubicles Technologies

Hyderabad, India

- Built offline-first React Native app for solar installations with WatermelonDB, facial recognition, and barcode scanning
- Developed oil industry CRM with integrated eSign and collaboration tools
- Created backend services with Django and Next.js with CI/CD pipelines

EDUCATION

JNTUH, Hyderabad, India

Bachelor of Technology, Electronics and Communication Engineering

2018 - 2022

CERTIFICATIONS

- [GitHub Foundations](#), GitHub (Mar 2025)
- [Microsoft AI](#), Microsoft (Jan 2021)
- [IBM Cloud Essentials](#), IBM (May 2021)
- [Stanford Internet of Things](#), Stanford (Jan 2021)

PROJECTS

Enterprise Conversational AI Platform

2025

- Production voice AI calling platform: 57 Rust WASM plugins (Extism PDK) across insurance, pharma, and diagnostics verticals; 5 telephony provider integrations with Temporal workflow orchestration; real-time STT/TTS pipeline with smart turn detection and LLM-based PII redaction; HMAC-SHA256 auth with replay prevention; multi-LLM orchestration across 10+ providers; multi-region AWS (us-east-1, ap-south-1) with Terraform/Terragrunt.

AroundMe - Location-Based Social Platform

2024

- Founded and architected distributed backend with FastAPI, PostgreSQL, Redis cluster, Elasticsearch for geospatial queries, Neo4j for social graph, and ML-powered recommendations. Led team of 10. Scaled to 65K+ downloads.

Memli - AI-Powered Flashcard App

2023

- React Native learning app with ML-generated mnemonics and YOLO computer vision to detect and explain textbook images. Built intelligent quiz system with personalized assessments and spaced repetition.

Rust CLI Tooling

2025

- Cargo workspace with 7 CLI tools for deployment automation and plugin management. Features Tokio async runtime, Google OAuth2 with token persistence, builder-pattern HTTP client, and unified error handling.